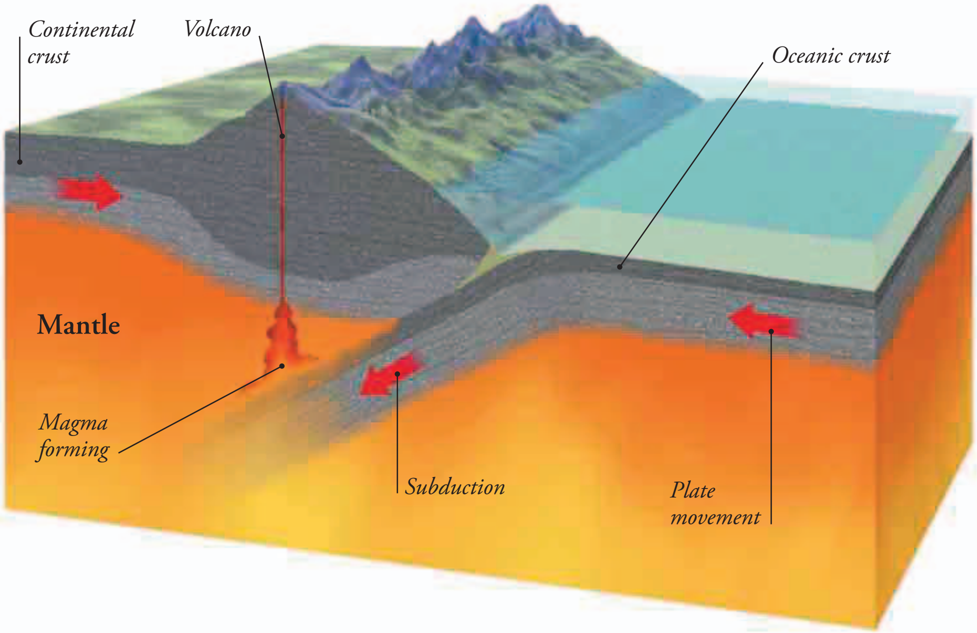
CONVERGENT BOUNDARIES

At a convergent boundary, where plates are pushing together, the edge of one plate can be subducted, or drawn beneath, the edge of the other. Oceanic crust is always dragged below continental crust because it is heavier. In places where two oceanic plates push together, the one with the oldest, heaviest rock is dragged beneath the newer one.



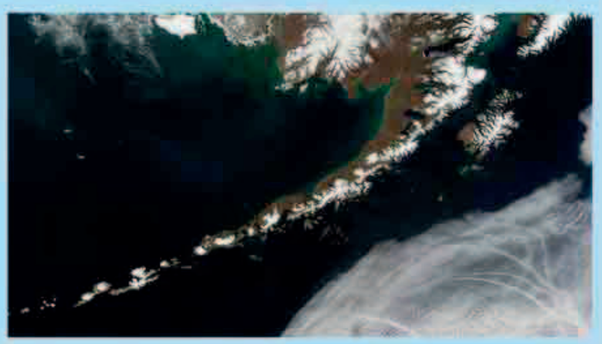
OCEAN TRENCHES

Oceanic subduction zones are marked by deep trenches in the ocean floor that follow the edge of the upper plate. On average, the ocean floors lie about 12,500 ft (3,800 m) below the surface, but some of these ocean trenches are at least twice as deep. The deepest part of the Mariana Trench in the western Pacific plunges to nearly 36,000 ft (11,000 m) below sea level, the deepest point on Earth's surface.

◀ DEEP BLUE
The dark blue line in this image of the northwest Pacific is the Kuril Trench, extending from off northern Japan to the Kamchatka Peninsula in Siberia.

VOLCANIC CHAINS

Subducted oceanic crust melts as it is carried down into the hot mantle. The melted rock may then erupt from volcanoes above the subduction zone. These form chains of volcanic islands along the plate boundary, known as island arcs, such as the Aleutian Islands off Alaska, shown in this satellite image.



SLIDING BOUNDARIES

At transform boundaries, the plates slide past each other without pulling apart or pushing together. This means that there is no new crust created, and no old crust destroyed. Most transform boundaries are on ocean floors, where they allow sections of oceanic crust to move at different rates. But some occur on land—the most famous example is the San Andreas Fault, which runs through California.

▶ FRACTURED LANDSCAPE
Moving plates on either side of the San Andreas Fault have ripped a long seam in the Carrizo Plain of California, and regularly cause earthquakes.